

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	# Customer ID	# Age	Gender	Item Purchased	Category	# Purchase Amount (USD)	Location
count	3900.0	3900.0	3900	3900	3900	3900.0	3900
unique	Missing value	Missing value	2	25	4	Missing value	50
top	Missing value	Missing value	Male	Blouse	Clothing	Missing value	Montana
freq	Missing value	Missing value	2652	171	1737	Missing value	96
mean	1950.5	44.06846153846154	Missing value	Missing value	Missing value	59.76435897435898	Missing value
std	1125.9773532358456	15.207589127162382	Missing value	Missing value	Missing value	23.685392250875307	Missing value
min	1.0	18.0	Missing value	Missing value	Missing value	20.0	Missing value
25%	975.75	31.0	Missing value	Missing value	Missing value	39.0	Missing value
50%	1950.5	44.0	Missing value	Missing value	Missing value	60.0	Missing value
75%	2925.25	57.0	Missing value	Missing value	Missing value	81.0	Missing value

11 rows x 18 cols

10 per page

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Size	Color	Season	#	Review Rating	Subscription Status	Shipping Type	Discount Applied
3900	3900	3900		3863.0	3900	3900	3900
4	25	4		Missing value	2	6	2
M	Olive	Spring		Missing value	No	Free Shipping	No
1755	177	999		Missing value	2847	675	2223
Missing value	Missing value	Missing value		3.750064716541548	Missing value	Missing value	Missing value
Missing value	Missing value	Missing value		0.7169829842073633	Missing value	Missing value	Missing value
Missing value	Missing value	Missing value		2.5	Missing value	Missing value	Missing value
Missing value	Missing value	Missing value		3.1	Missing value	Missing value	Missing value
Missing value	Missing value	Missing value		3.8	Missing value	Missing value	Missing value
Missing value	Missing value	Missing value		4.4	Missing value	Missing value	Missing value

11 rows x 18 cols 10 per page

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Promo Code Used	# Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.0	3900	3900
Missing value	Missing value	6	7
Missing value	Missing value	PayPal	Every 3 Months
223	Missing value	677	584
Missing value	25.35153846153846	Missing value	Missing value
Missing value	14.447125170462309	Missing value	Missing value
Missing value	1.0	Missing value	Missing value
Missing value	13.0	Missing value	Missing value
Missing value	25.0	Missing value	Missing value
Missing value	38.0	Missing value	Missing value

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- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Female	75191
Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69
55	94
57	73
58	64
60	79
62	68

1 ▾
>
>>
Number of rows: 25 ▾
Filter rows:

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	Average Purchase Amount (USD)
Express	60.48
Standard	58.46

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend ▾ 2	total_revenue ▲ 1
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases

item_purchased	discount_rate ▾ 1
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
New	83
Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
3	Accessories	Belt	161
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
2	Outerwear	Coat	161
1	Outerwear	Jacket	163

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2518
Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	total_revenue ▾ 1
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.